THE LYAPUNOV EXPONENT OF A RANDOMLY SWITCHED PAIR OF STABLE PLANAR SYSTEMS IS UNIMODAL IN THE SWITCHING RATE

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Abstract. Let X_t solve $\dot{X}_t = A_{I_t} X_t$ in the plane, where $A_0 = \left(\begin{smallmatrix} -\alpha & 1 \\ 0 & -\alpha \end{smallmatrix}\right)$, $A_1 = \left(\begin{smallmatrix} -\alpha & 0 \\ -1 & -\alpha \end{smallmatrix}\right)$ and I_t switches between 0 and 1 at rate r. Both matrices are Hurwitz, yet for small α the system blows up for an intermediate range of rates. The top Lyapunov exponent has the form $G(r) - \alpha$, and Malrieu conjectured that G increases and then decreases, with a single maximum of height about 0.2, and tends to zero at both ends. We prove this. The maximum is attained at a single $r_* \in (0.226, 0.235)$, $G(r) = r \log(1/r)\,(1 + O(1/\log(1/r)))$ as $r \to 0$, and $G(r) \le 1/(2r) + (2/\pi)K_1(r)$ for all r. Consequently the set of rates for which the system is unstable is an open interval when $\alpha < G(r_*)$ and is empty otherwise. The proof writes G as a ratio of two one-dimensional integrals of an explicit kernel, settles $r \le 0.01$ and $r \ge 5$ with elementary and Bessel-function bounds, and covers $[0.01, 100]$ by interval arithmetic.

1. Introduction

Fix $\alpha > 0$ and $r > 0$ and consider the Markov process (X_t, I_t) on $\mathbb{R}^2 \times \{0, 1\}$ with generator (1.1)

$$Lf(x,i) = (A_i x)\cdot \nabla_x f(x,i) + r\big(f(x, 1-i) - f(x,i)\big), \qquad A_0 = \begin{pmatrix} -\alpha & 1 \\ 0 & -\alpha \end{pmatrix}, \quad A_1 = \begin{pmatrix} -\alpha & 0 \\ -1 & -\alpha \end{pmatrix}.$$

Every convex combination $pA_1 + (1-p)A_0$ has eigenvalues $-\alpha \pm i\sqrt{p(1-p)}$, so each frozen system is exponentially stable, but Lawley, Mattingly and Reed [2] showed that for suitable α the switched system is unstable for an intermediate range of rates. Write $U_t = X_t/|X_t|$. The pair (U_t, I_t) has a unique invariant law μ_r, and almost surely

$$\lim_{t\to\infty} \frac{1}{t} \log |X_t| = \sum_i \int \langle A_i u, u\rangle \, \mu_r(\mathrm{d}u, i) = G(r) - \alpha,$$

where G does not depend on α and has an explicit integral formula [4, Theorem 4.1]. Malrieu [4, Conjecture 4.3] conjectured from numerical experiments that G is unimodal: there is r_c with $G' > 0$ on $(0, r_c)$ and $G' < 0$ on (r_c, ∞), $G(r_c) \approx 0.2$, and $G(r) \to 0$ as $r \to 0$ and as $r \to \infty$. He noted that this would make the instability set $\{r > 0 : G(r) > \alpha\}$ an interval for $\alpha < G(r_c)$ and empty for $\alpha > G(r_c)$, and asked for a proof [4, Open question 2]. For a family of rotation-type pairs, Lagasquie [3] proves that the exponent tends to its frozen value at both ends of the rate axis.

Theorem 1.1. *G is continuously differentiable on $(0, \infty)$, and there is $r_* \in (0.226, 0.235)$ such that $G' > 0$ on $(0, r_*)$ and $G' < 0$ on (r_*, ∞). Moreover*

$$\frac{\log(1/r) - \gamma}{1 + 4r^2(1 + 2K_0(2r))} \;\le\; \frac{G(r)}{r} \;\le\; \log\frac{1}{r} + 2K_0(2 - 2r^2) \qquad (0 < r < 1),$$

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where γ is Euler's constant, and $G(r) \le \frac{1}{2r} + \frac{2}{\pi}K_1(r)$ for all $r > 0$. In particular $G(r) \to 0$ at both ends.

Numerically $G(r_*) = 0.1985$. The two-sided bound gives $G(r) = r\log(1/r)\,(1+O(1/\log(1/r)))$; the quadrature values are consistent with $G(r) = r\log(1/r) - \gamma r + O(r^3\log^2(1/r))$.

Corollary 1.2. *Let $\alpha > 0$. If $\alpha < G(r_*)$, there are $r_1 < r_* < r_2$ such that the Lyapunov exponent $G(r) - \alpha$ is positive, and hence $|X_t| \to \infty$ almost surely, exactly for $r \in (r_1, r_2)$, and negative, and hence $|X_t| \to 0$ almost surely, for $r \notin [r_1, r_2]$. If $\alpha \ge G(r_*)$, the exponent is negative for every $r \ne r_*$.*

This follows from the theorem, since a continuous function that increases strictly, then decreases strictly, and tends to 0 at both ends has interval superlevel sets.

Remark 1.3. In [4] the maximum is placed at $r_c \approx 4.6$, read from Figure 1 there. For the system (1.1) the maximum is at $r_* \approx 0.23$. A direct simulation of (1.1) confirms this: integrating the two shear flows exactly over independent exponential holding times of rate r, with 2×10^6 switches, gives $\frac{1}{T}\log|X_T| + \alpha = 0.1985$, 0.1081, 0.0272, 0.0065 at $r = 0.23$, 1, 4.6, 20, against $G = 0.1985$, 0.1083, 0.0269, 0.0062 from the formula of Section 2. The shape, the height and the two limits are as conjectured.

The proof has three parts. Section 2 rewrites G in the coordinate $x = \cot\theta$, uses a symmetry exchanging the two regimes to reduce to $x > 0$, and arrives at $G = N_+/D_+$ with

$$N_+(r) = \int_0^\infty \Big(\frac{x}{1+x^2} - \frac{J(x,r)}{x}\Big)\,\mathrm{d}x, \qquad D_+(r) = \int_0^\infty \Big(1 - J(x,r) + \frac{J(x,r)}{x^2}\Big)\,\mathrm{d}x, \tag{1.2}$$

$$J(x,r) = rx\int_0^1 e^{-r\varphi(x,s)}\,\mathrm{d}s, \qquad \varphi(x,s) = xs + \frac{s}{x(1-s)}. \tag{1.3}$$

Since $G' = (N_+'D_+ - N_+D_+')/D_+^2$, the sign of G' is the sign of $H_+ = N_+'D_+ - N_+D_+'$. Section 3 proves $H_+ > 0$ for $r \le 0.01$, Section 4 proves $H_+ < 0$ for $r \ge 5$, and Section 5 covers $[0.01, 100]$ by validated numerics.

2. Reduction to a ratio of integrals

Since $A_i = -\alpha I + N_i$ with $N_0 = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $N_1 = \left(\begin{smallmatrix} 0 & 0 \\ -1 & 0 \end{smallmatrix}\right)$, the direction U_t moves by the shears N_i alone, and $\langle A_i u, u\rangle = -\alpha + \langle N_i u, u\rangle$. The law μ_r is invariant under $u \mapsto -u$, so we may work on the projective line with the coordinate $x = u_1/u_2 \in \mathbb{R} \cup \{\infty\}$. In regime 0 the flow $\dot u = N_0 u$ gives $\dot x = 1$, in regime 1 the flow $\dot u = N_1 u$ gives $\dot x = x^2$, and

$$\langle N_0 u, u\rangle = \frac{x}{1+x^2}, \qquad \langle N_1 u, u\rangle = -\frac{x}{1+x^2}.$$

Let f_0, f_1 be the densities of the invariant law of (x, I), up to a common factor. They satisfy $-f_0' + r(f_1 - f_0) = 0$ and $-(x^2 f_1)' + r(f_0 - f_1) = 0$, so the flux $f_0 + x^2 f_1$ is constant and may be taken equal to 1. Then

$$G(r) = \frac{\int_{\mathbb{R}} (f_0 - f_1)\frac{x}{1+x^2}\,\mathrm{d}x}{\int_{\mathbb{R}} (f_0 + f_1)\,\mathrm{d}x}. \tag{2.1}$$

Symmetry. The map $\Phi(x, i) = (-1/x, 1 - i)$ conjugates the dynamics to itself: if $\dot x = 1$ then $y = -1/x$ has $\dot y = y^2$, and if $\dot x = x^2$ then $\dot y = 1$; the switching rates are equal. By uniqueness, Φ preserves the invariant law, that is $f_1(y) = f_0(-1/y)/y^2$. The growth rate is also preserved, since $-y/(1+y^2) = x/(1+x^2)$ for $y = -1/x$. The half-line $\{x > 0\}$ in regime i is carried onto $\{x < 0\}$ in regime $1 - i$, so both integrals in (2.1) are twice their restrictions to $x > 0$.

The density on $x > 0$. With $f_1 = (1 - f_0)/x^2$ the first equation reads $f_0' + r(1 + x^{-2})f_0 = r/x^2$. Its solutions on $(0, \infty)$ differ by multiples of $e^{-r(x-1/x)}$, which is not integrable at 0, so

$$f_0(x) = \int_0^x \frac{r}{y^2}\, e^{-r[(x-y)+(1/y-1/x)]}\,\mathrm{d}y = \frac{r}{x}\int_0^1 \frac{e^{-r\varphi(x,s)}}{(1-s)^2}\,\mathrm{d}s,$$

after the substitution $y = x(1-s)$. Since $\partial_s\varphi = x + 1/(x(1-s)^2)$ and φ increases from 0 to ∞, integrating $r\,\partial_s\varphi\, e^{-r\varphi}$ over $s \in (0,1)$ gives $1 = J + f_0$, that is $f_0 = 1 - J$ with J as in (1.3). Put $b = 1 + x^2$. Then $(f_0 - f_1)\frac{x}{b} = \frac{x}{b} - \frac{J}{x}$ and $f_0 + f_1 = 1 - J + J/x^2$, which with (2.1) gives $G = N_+/D_+$ with (1.2).

Lemma 2.1. *For $x, r > 0$:*

(a) $1 - e^{-rx} > J > 0$ and $J \le x^2/b$;
(b) $\partial_r J = r\int_0^1 \frac{s^2}{(1-s)^2} e^{-r\varphi}\,\mathrm{d}s > 0$.

Proof. (a) $\varphi > xs$ gives $J < rx\int_0^1 e^{-rxs}\,\mathrm{d}s = 1 - e^{-rx}$. Also $\varphi - sb/x = s^2/(x(1-s)) \ge 0$, so $J \le rx\int_0^\infty e^{-rsb/x}\,\mathrm{d}s = x^2/b$. (b) $\partial_r J = x\int_0^1 (1 - r\varphi)e^{-r\varphi}\,\mathrm{d}s$, and

$$(1 - r\varphi)e^{-r\varphi} = \partial_s\big(se^{-r\varphi}\big) + r\,(s\partial_s\varphi - \varphi)\,e^{-r\varphi}, \qquad s\partial_s\varphi - \varphi = \frac{s^2}{x(1-s)^2}.$$

The term $se^{-r\varphi}$ vanishes at $s = 0$ and as $s \to 1$. □

The integrands of (1.2) are nonnegative and integrable by Lemma 2.1(a) and Lemma 4.1(a) below. For $r \in [r_1, r_2] \subset (0,\infty)$, Lemma 2.1(b) gives $0 < \partial_r J(x,r) \le (r_2/r_1)\,\partial_r J(x,r_1)$, and $\int_0^\infty (1 + x^{-2})\,\partial_r J(x,r_1)\,\mathrm{d}x = r_1^{-2}$ by (3.2) below. This dominates the r-derivatives $-\partial_r J/x$ and $\partial_r J\,(x^{-2} - 1)$ of the integrands, so N_+ and D_+ are C^1 and may be differentiated under the integral sign. Hence $G \in C^1(0,\infty)$ and $\operatorname{sign} G' = \operatorname{sign} H_+$, with

$$H_+ = N_+'D_+ - N_+D_+', \qquad N_+' = -\int_0^\infty \frac{\partial_r J}{x}\,\mathrm{d}x, \qquad D_+' = \int_0^\infty \partial_r J\Big(\frac{1}{x^2} - 1\Big)\,\mathrm{d}x. \tag{2.2}$$

3. Small rates

The x-integrals against $e^{-r\varphi} = e^{-rs(x+a/x)}$, $a = 1/(1-s)$, are Bessel functions:

$$\int_0^\infty x^{\nu-1}e^{-\beta(x+a/x)}\,\mathrm{d}x = 2a^{\nu/2}K_\nu(2\beta\sqrt{a}) \qquad (\beta, a > 0), \tag{3.1}$$

here with $\beta = rs$ and $2\beta\sqrt{a} = 2rs/\sqrt{1-s}$. Put $u = s/\sqrt{1-s}$, which increases from 0 to ∞, with $\mathrm{d}u = \frac{2-s}{2(1-s)^{3/2}}\,\mathrm{d}s$, and write $w = 1 - s$, so that $u^2w = (1-w)^2$, $0 < w \le 1$, $w \le u^{-2}$ and $\sqrt{w} = (1-w)/u$. We use the Mellin formula $\int_0^\infty v^{\mu-1}K_\nu(v)\,\mathrm{d}v = 2^{\mu-2}\Gamma(\frac{\mu-\nu}{2})\Gamma(\frac{\mu+\nu}{2})$, the bound $vK_1(v) \le 1$, and $K_0' = -K_1$; the last two give $K_0(v) \le K_0(y) + \log(y/v)$ for $0 < v \le y$, hence $\int_0^y K_0 \le y(1 + K_0(y))$.

Let $L = \log(1/r)$ and define $e_1, \dots, e_4$ by

$$rD = 2 + e_1, \quad N = 2L - 2\gamma + e_2, \quad r^2D' = -2 + e_3, \quad rN' = -2 + e_4,$$

where $N = 2N_+$ and $D = 2D_+$.

Lemma 3.1. *For every $r > 0$, $e_1, e_2, e_3, e_4 > 0$, and*

$$e_1 \le 8r^2\big(1 + 2K_0(2r)\big), \qquad e_3 \le 4r^2\big(1 + 2K_0(2\sqrt{2}\,r)\big), \qquad N \le 2L + 4K_0(2 - 2r^2) \quad (r < 1).$$

Proof. e_1. By Fubini and (3.1) with $\nu = 0$, $\int_0^\infty (1-J)\,\mathrm{d}x = \int_0^1 \frac{2r}{(1-s)^2} K_0\,\mathrm{d}s$ and $\int_0^\infty Jx^{-2}\,\mathrm{d}x = \int_0^1 2rK_0\,\mathrm{d}s$, the argument of K_0 being $2rs/\sqrt{1-s} = 2ru$. In the variable u,

$$D_+ = 4r\int_0^\infty u\,\frac{1+w^2}{1-w^2}\,K_0(2ru)\,\mathrm{d}u.$$

Since $4r\int_0^\infty uK_0(2ru)\,\mathrm{d}u = 1/r$, we get $e_1 = 2rD_+ - 2 = 16r^2\int_0^\infty \frac{uw^2}{1-w^2}K_0(2ru)\,\mathrm{d}u > 0$. Using $1-w = u\sqrt{w}$, $\frac{uw^2}{1-w^2} = \frac{w^{3/2}}{1+w} \le \min(\frac12, u^{-3})$, so

$$e_1 \le 16r^2\Big(\frac{1}{4r}\int_0^{2r} K_0 + \frac12 K_0(2r)\Big) \le 8r^2(1+2K_0(2r)).$$

e_2. By Lemma 2.1(a), $N_+ > \int_0^\infty (e^{-rx} - \frac{1}{1+x^2})\,\frac{\mathrm{d}x}{x} = L - \gamma$. For the upper bound take $\delta = r^2$; on $s \le 1-\delta$, $\varphi \le s(x + 1/(x\delta))$, so

$$J \ge \frac{\delta x^2}{1+\delta x^2}\Big(1 - e^{-r(1-\delta)(x+1/(x\delta))}\Big),$$

and

$$N_+ \le \int_0^\infty \Big(\frac{x}{1+x^2} - \frac{\delta x}{1+\delta x^2}\Big)\,\mathrm{d}x + \int_0^\infty e^{-r(1-\delta)(x+1/(x\delta))}\,\frac{\mathrm{d}x}{x} = \frac12\log\frac{1}{\delta} + 2K_0\big(2(1-\delta)r/\sqrt{\delta}\big),$$

which is $L + 2K_0(2-2r^2)$.

e_3. By (2.2) and Lemma 2.1(b), with (3.1) at $\nu = \pm 1$,

$$\int_0^\infty \Big(1+\frac{1}{x^2}\Big)\partial_r J\,\mathrm{d}x = 4r\int_0^\infty u^2 K_1(2ru)\,\mathrm{d}u = \frac{1}{r^2}, \tag{3.2}$$

$$I_1 := \int_0^\infty \frac{\partial_r J}{x^2}\,\mathrm{d}x = 4r\int_0^\infty \frac{u^2 w}{1+w}K_1(2ru)\,\mathrm{d}u.$$

Hence $D_+' = 2I_1 - 1/r^2$ and $e_3 = 4r^2 I_1 > 0$. With $\frac{w}{1+w} \le \min(\frac12, u^{-2})$, splitting at $u = \sqrt{2}$ and using $u^2K_1(2ru) \le u/(2r)$ on the first piece, $I_1 \le 1 + 2K_0(2\sqrt{2}\,r)$.

e_4. By (3.1) at $\nu = 0$, $M := \int_0^\infty \partial_r J\, x^{-1}\,\mathrm{d}x = 4r\int_0^\infty \frac{u^2\sqrt{w}}{1+w}K_0(2ru)\,\mathrm{d}u$, and $u^2\sqrt{w}/(1+w) = u - 2uw/(1+w)$. Since $4r\int_0^\infty uK_0(2ru)\,\mathrm{d}u = 1/r$, $e_4 = 2 - 2rM = 16r^2\int_0^\infty \frac{uw}{1+w}K_0(2ru)\,\mathrm{d}u > 0$. □

Proposition 3.2. *$H_+ > 0$ for $0 < r \le 0.01$.*

Proof. Since $H = N'D - ND' = 4H_+$, expanding gives exactly

$$\frac{r^2 H}{4} = (L-\gamma-1) + \frac{2e_2 + 2e_4 + e_1e_4 - 2e_1 - e_3N}{4} \;\ge\; (L-\gamma-1) - \frac{e_1}{2} - \frac{e_3N}{4},$$

by Lemma 3.1. For $r \le 0.01$, $K_0(y) \le K_0(1) + \log(1/y)$ with $K_0(1) < 0.4211$, and $4K_0(2-2r^2) < 0.456$, so Lemma 3.1 gives $e_1/2 \le 4r^2(2L + 0.456)$, $e_3 \le 8r^2 L$ and $N \le 2L + 0.456$. Hence

$$\frac{e_1}{2} + \frac{e_3 N}{4} \le e^{-2L}(2L + 0.456)(2L+4),$$

which is decreasing for $L \ge \log 100$ and at most 0.0128 there, while $L - \gamma - 1 \ge 3.027$. □

The lower bound in Theorem 1.1 for small r is $G = N/D \ge r(2L - 2\gamma)/(2 + e_1)$, and the upper bound is $G \le rN/2$; both are from Lemma 3.1.

4. Large rates

Lemma 4.1. *For $x, r > 0$:*

(a) $0 \le N_+ \le \bar N := \frac{\pi}{4r} + K_1(r)$ *and* $D_+ \ge \frac{\pi}{2}$*;*

(b) $\partial_r J \le \frac{16}{r^2x^3} + xe^{-rx/2}$*, so* $\int_X^\infty \partial_r J\,\mathrm{d}x \le \frac{8}{r^2X^2} + e^{-rX/2}\big(\frac{2X}{r} + \frac{4}{r^2}\big)$*;*

(c) $\int_{1/2}^1 \partial_r J\,\mathrm{d}x \ge 0.016\,(1-Q)/r^2$*, where* $Q = e^{-a}(1 + a + a^2/2)$*,* $a = 5r/2$*.*

Proof. (a) The integrand of N_+ is $\frac{1}{xb}(x^2 - bJ) \ge 0$ by Lemma 2.1(a). Write

$$\frac{x^2}{b} - J = rx\int_0^1 e^{-rsb/x}\Big(1 - e^{-rs^2/(x(1-s))}\Big)\,\mathrm{d}s + rx\int_1^\infty e^{-rsb/x}\,\mathrm{d}s.$$

On $s \le \frac12$ use $1 - e^{-c} \le c \le 2rs^2/x$; the rest is at most $rx\int_{1/2}^\infty e^{-rsb/x}\,\mathrm{d}s = \frac{x^2}{b}e^{-rb/(2x)}$. So $x^2 - bJ \le \frac{4x^3}{rb^2} + x^2e^{-rb/(2x)}$. Dividing by xb, using $x/b \le \frac12$, $\int_0^\infty x^2b^{-3}\,\mathrm{d}x = \pi/16$ and $\int_0^\infty e^{-(r/2)(x+1/x)}\,\mathrm{d}x = 2K_1(r)$ gives the bound on N_+. Since $J \le x^2/b$, $1 - J \ge 1/b$, so $D_+ \ge \int_0^\infty b^{-1}\,\mathrm{d}x = \pi/2$.

(b) With $t = s/(1-s)$, $\varphi = \frac{xt}{1+t} + \frac{t}{x}$ and $\partial_r J = r\int_0^\infty \frac{t^2}{(1+t)^2}e^{-r\varphi}\,\mathrm{d}t$. For $t \le 1$, $\frac{xt}{1+t} \ge \frac{xt}{2}$; for $t \ge 1$, $\frac{xt}{1+t} \ge \frac{x}{2}$ and $\frac{t^2}{(1+t)^2} \le 1$. So $\partial_r J \le r\int_0^\infty t^2e^{-rxt/2}\,\mathrm{d}t + re^{-rx/2}\int_0^\infty e^{-rt/x}\,\mathrm{d}t$.

(c) For $x \in [\frac12, 1]$ and $t \in [0,1]$, $\varphi \le t(x + 1/x) \le \frac52 t$ and $\frac{t^2}{(1+t)^2} \ge \frac{t^2}{4}$, so $\partial_r J \ge \frac{r}{4}\int_0^1 t^2e^{-5rt/2}\,\mathrm{d}t = \frac{4(1-Q)}{125\,r^2}$. □

Proposition 4.2. $H_+ < 0$ *for* $r \ge 5$.

Proof. By (2.2),

$$-H_+ = \int_0^\infty \partial_r J\,\Big[\frac{D_+}{x} + \frac{N_+}{x^2} - N_+\Big]\,\mathrm{d}x.$$

The bracket is at least $D_+/x - N_+$, so it is negative only for $x > D_+/N_+ \ge X := (\pi/2)/\bar N$, and there it is at least $-\bar N$. On $[\frac12, 1]$ it is at least $\frac{\pi}{2} - \bar N$. For $r \ge 5$, $\bar N < 0.17$ and $X > 9$, so by Lemma 4.1

$$-r^2H_+ \;\ge\; 0.016\,(1-Q)\Big(\frac{\pi}{2} - \bar N\Big) - \bar N\Big(\frac{8}{X^2} + e^{-rX/2}(2rX + 4)\Big) =: \ell(r).$$

As r grows, $\bar N$ and Q decrease, X and $rX = \frac{\pi}{2}\,r^2/(\frac{\pi}{4} + rK_1(r))$ increase, and $v \mapsto e^{-v/2}(2v+4)$ is decreasing; so ℓ is increasing. Finally $\ell(5) > 0.0089$. □

With Lemma 4.1(a), $G = N_+/D_+ \le 2\bar N/\pi = \frac{1}{2r} + \frac{2}{\pi}K_1(r)$.

5. The middle range

On $[0.01, 100]$ the sign of G' is certified with ball arithmetic in Arb [1] through python-flint, at 64 bits of precision. These computations use the formula (2.1) on the whole line, without the symmetry, with f_0 written as an integral over $u \in (0,\infty)$ for $x > 0$ and over a finite u-interval for $x < 0$, so they also check the reduction of Section 2.

For an interval $R = [r_{\mathrm{lo}}, r_{\mathrm{hi}}]$ the parameter enters as a ball, so every enclosure holds for all $r \in R$ at once. The x-integrals of N, D and of their first and second r-derivatives are split into three parts.

- A strip $|x| \le \varepsilon$ with $\varepsilon = 10^{-5}\min(r_{\mathrm{lo}}, 1)$. The integrand is smooth but not analytic at $x = 0$, since the branch $x < 0$ carries $e^{-r(1+x^2)/|x|}$; it is bounded directly (for instance $0 \le d \le 2(1+x^2)x^3/r$ for $x > 0$, where $d = (1+x^2)f_0 - 1$).
- Tails $|x| \ge X$ with $X = 10^4/\min(r_{\mathrm{lo}}, 1)$ (and $10^5/\min(r_{\mathrm{lo}}, 1)$ for the derivatives), bounded by inequalities of the type of Lemma 4.1(b).

- Graded panels $[a, 2a]$ on each side between ε and X. On each panel the integral is enclosed by the 24-point Gauss–Legendre rule plus the remainder bound $\frac{64}{15}\, h\, M\rho^{-48}/(\rho^2-1)$ for the Bernstein ellipse E_ρ with $\rho = 1.6$ [5, Theorem 19.3], where h is the half-length and M bounds the integrand on the boundary of E_ρ (maximum modulus), evaluated on 48 boxes covering it. On these ellipses $|\arg x| < \pi/4$, which keeps the complex continuation of the integrand analytic; its value at a complex x is itself a rigorous Arb integral in u, built from exp, expm1 and division only.

Every strip and tail bound is a sum of powers of $1/r$ and terms e^{-cr}, $c > 0$, hence decreasing in r, and is evaluated at r_{lo}.

These give enclosures of N, D, N', D', hence of $G' = (N'D - ND')/D^2$, and of $H' = N''D - ND''$ (the terms $N'D'$ cancel). An interval on which the enclosure does not decide a sign is bisected. The results are in Table 1.

range of r	certified	intervals
$[0.01, 0.215]$	$G' > 0$	47, ratio 1.08, bisected
$[0.215, 0.25]$	$H' < 0$; $G' > 0$ on $[0.215, 0.226]$ and $G' < 0$ on $[0.235, 0.25]$	35, width 0.001
$[0.25, 100]$	$G' < 0$	83, ratio 1.08, bisected

TABLE 1. Certified signs. Each range is a contiguous chain of decided intervals.

On $[0.215, 0.25]$, H is strictly decreasing, so it vanishes at most once there. It is positive at 0.226 and negative at 0.235, so G' has exactly one zero r_* in $[0.215, 0.25]$, and $r_* \in (0.226, 0.235)$.

As an independent check of the code, fifteen decided intervals were recomputed with ordinary adaptive quadrature at both endpoints and the midpoint; the values of G and the central differences of G fell inside the stored enclosures in all 45 cases.

Proof of Theorem 1.1. By Proposition 3.2 and Table 1, $G' > 0$ on $(0, r_*)$; by Table 1 and Proposition 4.2, $G' < 0$ on (r_*, ∞). The bounds on G are those at the ends of Sections 3 and 4. □

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In November 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, found the proofs of Sections 2–4, wrote and ran the interval computations of Section 5 and the simulation

of Remark 1.3, and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code for the interval computations, the independent checks and the simulation, with their outputs, is available at
`https://github.com/yspennstate/malrieu-switched-lyapunov-unimodal`.

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